

Aditya Sahni

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EDUCATION

McMaster University

B.A.Sc. Honors Computer Science, Minor in Statistics

Hamilton, Ontario

Aug. 2022 – July 2027

EXPERIENCE

TD Bank

Software Engineer Co-op

Toronto, Ontario

Sept 2025 – Dec 2025

- Built an automated ETL pipeline with **Power Automate** to extract unstructured Confluence data and transform it into SQL-ready CSV formats, cutting manual data entry time by **90%**.
- Developed **SQL stored procedures** to process live TMX user requests, automating security categorizations (Blocklist, Watchlist, Safelist) and replacing a legacy manual tracking workflow.
- Reduced **5+ hours of weekly** manual work by writing **Python** validation scripts that enforce data integrity checks prior to database ingestion.
- Designed a **Power BI** dashboard tracking aggregator login trends and SLA metrics; tuned **DAX** queries to cut report refresh time.

DocOnline Health India Pvt. Ltd.

Assistant Software Developer

Mumbai, India

May 2024 – Sep 2024

- Built **REST APIs** in **Node.js** with a **PostgreSQL** backend for a medical data aggregation platform serving live patient data validation.
- Deployed a cloud backend on **AWS S3**, **Lambda**, and **EC2**, reducing operational infrastructure overhead by over **30%**.
- Integrated unit tests into the **CI/CD pipeline** with **GitHub Actions**, catching regressions earlier and accelerating feature delivery.

Axesouár Jewellery

Full-Stack Web Developer (Freelance)

Hamilton, Ontario

Jan 2025 – March 2025

- Built a **React.js** and **Node.js/MongoDB** e-commerce platform managing 120+ SKUs; optimized assets to improve Core Web Vitals (LCP) by **60%** and increase session retention.

PROJECTS

MapleRewards | Consumer Credit Card & Travel SaaS Platform | Go, Next.js, Redis, PostgreSQL

- Building Canada's first native credit card optimization platform (92 cards, 19 loyalty programs) with a freemium model targeting \$820B+ in annual Canadian card spending.
- Built a trip planner that scrapes live flight prices via **SerpAPI** and **Apify**, cross-references fares against 19 loyalty transfer matrices, and injects wallet context into an **LLM** to surface optimal point-vs-cash redemptions per route.
- 12 shipped features with **Go/Chi** backend, **PostgreSQL**, **Redis** caching, **Stripe** billing, **Google OAuth**, and 27,760 lines of production code.

Flux | Institutional-Grade Trading Platform | Go, PostgreSQL, Redis, Next.js, Docker

- Architected a distributed microservices trading engine in **Go** with **goroutine concurrency** and **mutex locking**, aggregating disparate market streams into a unified **WebSocket** feed for sub-second synchronization.
- Implemented a financial-grade ledger using **PostgreSQL** row-level locking (**SELECT FOR UPDATE**) for strict **ACID** compliance, preventing race conditions and guaranteeing portfolio accuracy.
- Designed a high-performance **Next.js** dashboard with **GPU-accelerated candlestick charts** (HTML5 Canvas) and a hybrid **Redis** hot-data cache, deployed via **Docker** for production consistency.

SonicID | Shazam-Style Audio Fingerprinting Engine | Python, Go, Next.js, PostgreSQL

- Implemented the **Wang 2003** audio fingerprinting algorithm from scratch using **STFT**, peak-picking, and combinatorial hashing of anchor-target pairs.
- Polyglot stack: **Python FastAPI** for DSP, **Go/Chi** for the match path with **PostgreSQL** BTREE lookups and histogram voting; **sub-200ms p50 latency** on a 500-track index.
- Next.js** frontend captures raw PCM via **AudioWorkletNode** with client-side WAV encoding to bypass browser MediaRecorder issues.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, SQL, HTML/CSS

Frameworks: Next.js, React, Node.js, Chi, Gin, FastAPI, Tailwind CSS

Databases & Cloud: PostgreSQL, Redis, MongoDB, SQLite, AWS (EC2, S3, Lambda), Docker, Vercel

Tools & Practices: Git, GitHub Actions, REST APIs, WebSockets, JWT, CI/CD, Microservices